

Research Notes on Natural language processing

Natural language processing

>> Section 1

Natural language processing has its roots in the 1950s. Already in 1950, Alan Turing published an article titled "Computing Machinery and Intelligence," which proposed what is now called the Turing test as a criterion of intelligence, though at the time that was not articulated as a problem separate from artificial intelligence. The proposed test includes a task that involves the automated interpretation and generation of natural language.

>> Section 2

Relational semantics (semantics of individual sentences)
Relationship extraction Given a chunk of text, identify the relationships among named entities (e.g. who is married to whom). Semantic parsing Given a piece of text (typically a sentence), produce a formal representation of its semantics, either as a graph (e.g., in AMR parsing) or in accordance with a logical formalism (e.g., in DRT parsing). This challenge typically includes aspects of several more elementary NLP tasks from semantics (e.g., semantic role labelling, word-sense disambiguation) and can be extended to include full-fledged discourse analysis (e.g., discourse analysis, coreference; see Natural language understanding below). Semantic role labelling (see also implicit semantic role labelling below) Given a single sentence, identify and disambiguate semantic predicates (e.g., verbal frames), then identify and classify the frame elements (semantic roles).

>> Section 3

Interest on increasingly abstract, "cognitive" aspects of natural language (1999–2001: shallow parsing, 2002–03: named entity recognition, 2006–09/2017–18: dependency syntax, 2004–05/2008–09 semantic role labelling, 2011–12 coreference, 2015–16: discourse parsing, 2019: semantic parsing). Increasing interest in multilinguality, and, potentially, multimodality (English since 1999; Spanish, Dutch since 2002; German since 2003; Bulgarian, Danish, Japanese, Portuguese, Slovenian, Swedish, Turkish since 2006; Basque, Catalan, Chinese, Greek, Hungarian, Italian, Turkish since 2007; Czech since 2009; Arabic since 2012; 2017: 40+ languages; 2018: 60+/100+ languages) Elimination of symbolic representations (rule-based over supervised towards weakly supervised methods, representation learning and end-to-end systems)

>> Section 4

Natural language processing (NLP) is the processing of natural language information by a computer. NLP is a subfield of computer science and is closely associated with artificial intelligence. NLP is also related to information retrieval, knowledge representation, computational

linguistics, and linguistics more broadly. Major processing tasks in an NLP system include: speech recognition, text classification, natural language understanding, and natural language generation.

>> Section 5

both statistical and neural network methods tend to focus more on the most common cases extracted from a corpus of texts, whereas the rule-based approach needs to provide rules for both rare and common cases equally. language models, produced by either statistical or neural networks methods, are more robust to both unfamiliar (e.g. containing words or structures that have not been seen before) and erroneous input (e.g. with misspelled words or words accidentally omitted) in comparison to the rule-based systems, which are also more costly to produce. the larger such a (probabilistic) language model is, the more accurate it becomes, in contrast to rule-based systems that can gain accuracy only by increasing the amount and complexity of the rules leading to intractability problems. Rule-based systems are commonly used:

>> Section 6

Automatic summarization (text summarization) Produce a readable summary of a chunk of text. Often used to provide summaries of the text of a known type, such as research papers, articles in the financial section of a newspaper. Grammatical error correction Grammatical error detection and correction involves a great band-width of problems on all levels of linguistic analysis (phonology/orthography, morphology, syntax, semantics, pragmatics). Grammatical error correction is impactful since it affects hundreds of millions of people that use or acquire English as a second language. It has thus been subject to a number of shared tasks since 2011. As far as orthography, morphology, syntax and certain aspects of semantics are concerned, and due to the development of powerful neural language models such as GPT-2, this can now (2019) be considered a largely solved problem and is being marketed in various commercial applications. Logic translation Translate a text from a natural language into formal logic. Machine translation (MT) Automatically translate text from one human language to another. This is one of the most difficult problems, and is a member of a class of problems colloquially termed "AI-complete", i.e. requiring all of the different types of knowledge that humans possess (grammar, semantics, facts about the real world, etc.) to solve properly. Natural language understanding (NLU) Convert chunks of text into more formal representations such as first-order logic structures that are easier for computer programs to manipulate. Natural language understanding involves the identification of the intended semantic from the multiple possible

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semantics which can be derived from a natural language expression which usually takes the form of organized notations of natural language concepts. Introduction and creation of language metamodel and ontology are efficient however empirical solutions. An explicit formalization of natural language semantics without confusions with implicit assumptions such as closed-world assumption (CWA) vs. open-world assumption, or subjective Yes/No vs. objective True/False is expected for the construction of a basis of semantics formalization. Natural language generation (NLG): Convert information from computer databases or semantic intents into readable human language. Book generation Not an NLP task proper but an extension of natural language generation and other NLP tasks is the creation of full-fledged books. The first machine-generated book was created by a rule-based system in 1984 (Racter, The policeman's beard is half-constructed). The first published work by a neural network was published in 2018, 1 the Road, marketed as a novel, contains sixty million words. Both these systems are basically elaborate but non-sensical (semantics-free) language models. The first machine-generated science book was published in 2019 (Beta Writer, Lithium-Ion Batteries, Springer, Cham). Unlike Racter and 1 the Road, this is grounded on factual knowledge and based on text summarization. Document AI A Document AI platform sits on top of the NLP technology enabling users with no prior experience of artificial intelligence, machine learning or NLP to quickly train a computer to extract the specific data they need from different document types. NLP-powered Document AI enables non-technical teams to quickly access information hidden in documents, for example, lawyers, business analysts and accountants. Dialogue management Computer systems intended to converse with a human. Question answering Given a human-language question, determine its answer. Typical questions have a specific right answer (such as "What is the capital of Canada?"), but sometimes open-ended questions are also considered (such as "What is the meaning of life?"). Text-to-image generation Given a description of an image, generate an image that matches the description. Text-to-scene generation Given a description of a scene, generate a 3D model of the scene. Text-to-video Given a description of a video, generate a video that matches the description.